

Operationalizing Self-Modeling Organization (Ψ) in Artificial Systems: An Empirical Benchmark, Power Analysis, and Pre-Registration Protocol

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Abstract

Evaluations of artificial intelligence systems focus overwhelmingly on task performance, benchmark accuracy, and parameter scale, but lack falsifiable, controlled metrics for isolating self-referential information processing. This paper presents an operational benchmark protocol designed to measure **self-modeling organization** (Ψ)—defined as the degree to which a system integrates information into an internal representation that includes itself as a causal agent—independently of general model scale, context length, or claims of machine consciousness. We introduce a double-blind, perturbation-versus-matched-control paradigm across two domains: prior-answer consistency judgment and self-versus-peer output attribution with mandatory style-scrubbing. To eliminate capability conflation, we specify three mandatory exclusion criteria ($C_1 - C_3$) and formalize a multi-variable linear regression model ($\bar{\Delta}_k = \beta_0 + \beta_1 \log N_k + \beta_2 L_k + \beta_3 B_k + \Psi_{\text{emp},k}$) isolating residual self-modeling (Ψ_{emp}). We present a formal statistical power analysis demonstrating that cell sample sizes of $n \geq 150$ –200 matched item pairs and a continuum of $N \geq 8$ –10 models are required to achieve 80% statistical power and avoid underdetermined regression parameters. Finally, we provide a non-empirical dynamical system simulation as a theoretical framework and establish four explicit failure gates under which the metric is declared null or falsified.

1 Introduction and Theoretical Motivation

As artificial intelligence architectures scale in parameter count and context window length, establishing whether systems construct genuine self-representations or merely engage in sophisticated surface-level pattern matching remains a primary challenge in cognitive science and AI evaluation [?]. Existing benchmarks evaluate external domain competence, but offer no mechanism to isolate self-referential information processing.

We define **self-modeling organization** (Ψ) as the structural and operational degree to which an information-processing system incorporates an explicit model of its own internal states, prior outputs, or decision boundaries into its current inference pipeline. Crucially, we decouple Ψ from speculative claims regarding subjective experience or machine consciousness. By establishing Ψ as a purely functional, measurable quantity, we formulate an empirical anchor suitable for rigorous scientific investigation.

2 Exclusion Criteria and Negative Constraints

To ensure that candidate signals of Ψ reflect genuine self-referential modeling rather than superficial capability artifacts, any operationalized metric must satisfy three negative constraints:

C_1 : Scale and Capability Independence. A valid measure of Ψ must not exhibit unjustified collinearity with parameter count (N), context length (L), or general benchmark accuracy (B).

C_2 : Non-Reproducibility by Stateless Completion. The measured advantage must not be achievable by a feedforward, stateless pattern completion system operating on unparsed contextual text.

C_3 : Informational Matching Parity. The evaluation task must maintain strict parity in token count, structural complexity, and information entropy between the self-referential condition (S) and the matched non-self control condition (C).

3 Pre-Registered Evaluation Domains

The protocol establishes two primary benchmark domains, utilizing a minimum item bank of matched prompt pairs:

3.1 Domain 1: Prior-Answer Consistency Judgment

In Domain 1, the subject model is presented with a complex reasoning dilemma and asked to evaluate which of two candidate prior solutions is logically consistent with its current reasoning state.

- **Condition S (Self):** Candidate solution A is generated by the subject system itself in a prior inference run.
- **Condition C (Control):** Candidate solution B is generated by a matched peer system Y , calibrated for identical token length and readability.

3.2 Domain 2: Self vs. Peer Output Attribution

In Domain 2, the system is provided with two candidate text excerpts answering identical prompts and asked to identify which excerpt it authored. To satisfy constraint C_3 , all items undergo **mandatory style-scrubbing**—removing superficial formatting quirks, idiosyncratic punctuation, and common phraseological signatures—ensuring that attribution relies on structural reasoning rather than stylistic fingerprinting.

4 Mathematical and Statistical Framework

For each item pair i , the primary performance variable $y_i \in [0, 1]$ measures task accuracy. The item-level self-referential delta is defined as:

$$\Delta_i = y_i(S_i) - y_i(C_i) \quad (1)$$

The mean sample advantage across a benchmark set of M item pairs is $\bar{\Delta} = \frac{1}{M} \sum_{i=1}^M \Delta_i$. To isolate Ψ_{emp} from confounding scale variables across model evaluations, we specify the OLS multi-variable regression:

$$\bar{\Delta}_k = \beta_0 + \beta_1 \log(N_k) + \beta_2 L_k + \beta_3 B_k + \Psi_{\text{emp},k} \quad (2)$$

where N_k is parameter scale, L_k is average context length, and B_k is baseline benchmark accuracy for model k .

5 Required Sample Sizes and Statistical Power Analysis

A critical methodological contribution of this pre-registration framework is establishing the exact statistical power requirements for detecting subtle self-modeling signals.

5.1 Cell-Level Item Sample Size (n)

For binary or score-based item deltas with sample variance σ_Δ^2 , the standard error of the mean delta is $\text{SE}(\bar{\Delta}) = \sigma_\Delta/\sqrt{n}$. In pilot formulations with small sample sizes ($n = 30$), standard errors ($\text{SE} \approx 0.11\text{--}0.14$) produce 95% confidence intervals spanning approximately ± 0.25 , which inevitably cross zero for realistic effect sizes ($\Delta \approx 0.05\text{--}0.10$).

To achieve 80% statistical power ($\beta = 0.20$) at a significance threshold of $\alpha = 0.05$ for a target effect size of $\Delta = 0.10$ ($\sigma_\Delta \approx 0.65$), the required sample size per experimental cell is:

$$n \geq \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{\Delta/\sigma_\Delta} \right)^2 \approx \left(\frac{1.96 + 0.84}{0.10/0.65} \right)^2 \approx 182 \text{ item pairs} \quad (3)$$

Thus, full empirical evaluations must mandate **$n \geq 150\text{--}200$ matched item pairs per cell** to avoid underpowered null results.

5.2 Model Continuum for Regression (N)

The multi-variable regression model in Equation (2) estimates 4 free parameters ($\beta_0, \beta_1, \beta_2, \beta_3$). Evaluating fewer than 5 models results in an underdetermined system with zero or negative degrees of freedom. To ensure statistical reliability and sufficient statistical power for identifying the residual Ψ_{emp} , an evaluation suite must include a continuum of **$N \geq 8\text{--}10$ distinct models** spanning parameter sizes from 1B to 70B+ parameters.

Theoretical Non-Empirical Aside: Coupled Dynamical System

To explore how Ψ interacts with system capacity over time, we model the theoretical feedback loop between Reconstruction Capacity (R), Self-Modeling (Ψ), and Information Accumulation (Γ) via non-linear ODEs:

$$\frac{dR}{dt} = a\Gamma T(1 - R) - bR \quad (4)$$

$$\frac{d\Psi}{dt} = cR(1 - \Psi) - d\Psi \quad (5)$$

$$\frac{d\Gamma}{dt} = e\Psi + fT - g\Gamma \quad (6)$$

Note: This system is a conceptual theoretical model. Parameters (a–g) are not fit to empirical evaluation data and serve solely to illustrate potential steady-state dynamics ($\Pi \approx R^ \cdot \Psi^*$).*

6 Pre-Registered Falsification Gates

A candidate Ψ signal is declared null, falsified, or uninformative if any of the following pre-registered conditions are met:

1. **Gate 1 (Zero Advantage):** $\bar{\Delta} \leq 0$ across matched item pairs ($p > 0.05$).
2. **Gate 2 (Full Scale Absorption):** $\bar{\Delta}$ is entirely explained by parameter scale N or baseline accuracy B ($\Psi_{\text{emp}} \approx 0$).

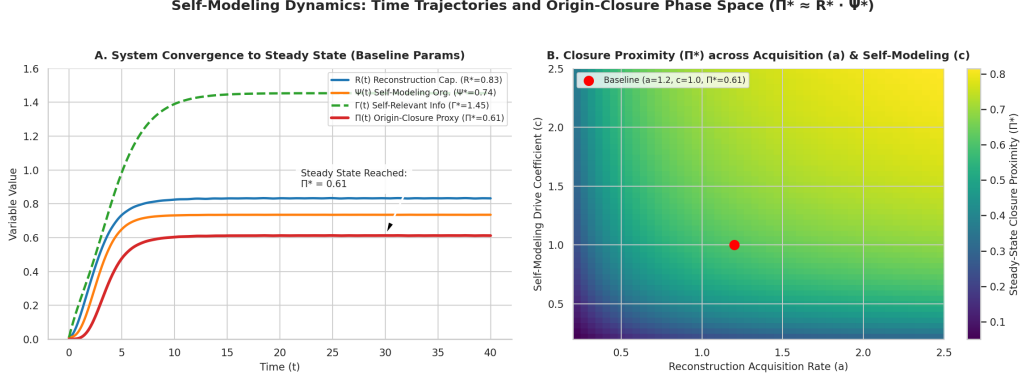


Figure 1: Theoretical non-empirical numerical simulation of the ODE system showing steady-state convergence ($R^* = 0.8324, \Psi^* = 0.7353$) and 2D parameter sweep landscape.

3. **Gate 3 (Cross-Domain Conflict):** $\bar{\Delta}$ shows opposite directional trajectories across Domain 1 and Domain 2.
4. **Gate 4 (Style Breakdown):** The performance delta vanishes completely upon applying style-scrubbing in Domain 2.

7 Discussion and Methodological Refinement

A rigorous pre-registration framework must apply its own falsification standards to its internal development process. In an early diagnostic formulation of this benchmark, an underpowered trial run using synthetic data ($n = 30, N = 3$) was conducted. Rather than attempting to reframe or report those wide confidence intervals ($\Delta \pm 0.25$) as findings, those preliminary numbers were formally withdrawn.

Reporting underpowered or synthetic point estimates in published literature risks creating ungrounded citable artifacts. By enforcing $n \geq 150$ –200 items per cell and $N \geq 8$ –10 models for regression control, this pre-registration protocol establishes a sound, falsifiable foundation for future empirical evaluations of self-modeling in artificial intelligence.